

OKAFOR CHIBUZOR AUTHUR

AI Engineer | LLM Systems | RAG Architectures

Lagos, Nigeria | [Phone](#) | [Email](#) | [GitHub](#) | [LinkedIn](#) | [Portfolio](#) | [Substack](#)

PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer specializing in large language model (LLM) systems, Retrieval-Augmented Generation (RAG), and multi-agent AI architectures. Experienced in designing and shipping production-grade AI platforms end-to-end using Go and Python backends with FastAPI and Chi, paired with React and Next.js frontends, and scalable vector search systems built on PostgreSQL with pgvector and Docker-orchestrated infrastructure. Proven ability to architect and deliver complex AI workflows including document retrieval, email intelligence and automation, speaker-attributed transcription, persistent conversational agents, real-time communication systems, and subscription billing systems, consistently building complete multi-tenant SaaS products rather than isolated features.

Built and shipped multiple production systems including [LearnVault](#), a multi-tenant AI learning and collaboration platform; [MailChecker](#), a centralized AI-powered email intelligence system; and InfiniMail, a full-stack email automation and workflow platform with authentication, billing, and campaign orchestration. Strong focus on system design, scalability, and production reliability across distributed AI workloads.

Strong written English (C1+) with the ability to produce clear technical documentation and communicate complex systems effectively. Focused on building impactful AI products that automate complex workflows with measurable value from large-scale data.

CORE SKILLS

- **AI Systems:** LLMs, RAG, Multi-Agent Architectures, Prompt Engineering, LangChain, LangGraph, HuggingFace Transformers, OpenAI APIs
- **Backend:** FastAPI, PostgreSQL (pgvector), Redis, Celery, Golang, Python
- **Infrastructure:** Docker, Nginx, AWS, GCP, DigitalOcean, Caddy
- **Data & ML:** Python, Pandas, Scikit-Learn, FAISS, NLP (SpaCy, NLTK, Gensim)

PROFESSIONAL EXPERIENCE

AI Engineer (Contract)

[Kuuka](#)

Remote | 2025 – 2026

Technologies: Python, FastAPI, Docker, PostgreSQL, Redis, Celery, CI/CD, RBAC, Microservices Architecture

- Built AI-driven automation systems for supplier intelligence across thousands of vendor records, improving procurement decision workflows.
- Designed multi-channel campaign and notification platform supporting segmentation, analytics, templates, and automated delivery pipelines across tenants
- Integrated enterprise tools including financial, analytics, and messaging platforms to enable AI agents to interact with operational business systems
- Improved deployment stability by hardening Docker builds and resolving service orchestration failures across distributed Services
- Implemented CI-ready automated test coverage across 50+ API endpoints, improving deployment reliability and reducing runtime failures
- Designed audit logging, incident monitoring, GDPR export flows, and RBAC-based tenant isolation to strengthen governance and compliance infrastructure

AI Engineer (Contract)

[Possibl](#)

Remote | 2025

Technologies: Python, FastAPI, LangChain, LangGraph, PostgreSQL, SQL, Async APIs, Data Visualization, Docker

- Architected multi-agent conversational AI platform using FastAPI, LangChain, and LangGraph for intent routing and dynamic workflow orchestration, enabling 3+ coordinated agents to process natural-language queries and execute multi-step data retrieval workflows.
- Designed natural-language analytics pipeline translating user queries into SQL execution workflows, enabling automated data exploration and visualization generation
- Implemented PostgreSQL-based checkpointed graph execution with automated memory summarization to maintain persistent conversational context
- Designed human-in-the-loop decision agents for table selection and context sufficiency validation before workflow execution
- Refactored asynchronous API infrastructure with structured outputs and resilient error handling to improve system reliability

Applied AI Engineer (Contract)

[Zenovo AI](#)

Remote | 2024 – 2025

Technologies: FastAPI, LangChain, PostgreSQL (pgvector), Redis, Celery, Docker

- Designed AI-powered proposal generation platform automating RFQ and RFP response workflows using multi-agent LLM orchestration, enabling automated analysis of 100+ page procurement documents and reducing proposal preparation time by 70%.
- Architected Retrieval-Augmented Generation (RAG) platform using PostgreSQL and pgvector enabling semantic retrieval across large document repositories and supporting scalable AI-driven knowledge workflows
- Engineered scalable document ingestion pipeline supporting PDF parsing, semantic chunking, and cloud synchronization to enable automated indexing for AI knowledge retrieval systems
- Engineered asynchronous processing infrastructure using Redis caching and Celery workers for scalable AI task execution
- Deployed containerized microservices with Docker and automated CI/CD pipelines supporting reliable production deployments

AI Engineer / Machine Learning Consultant

Independent Clients

2024 – 2025

- Developed Retrieval-Augmented Generation knowledge retrieval system using FAISS vector indexing for large document Datasets
- Built speech-to-text transcription API using Whisper large model deployed on Linux infrastructure
- Designed AI-powered data exploration tool enabling analysts to query Excel datasets using natural language
- Implemented automated document embedding and indexing pipelines to support scalable knowledge base retrieval

AI SYSTEMS & ARCHITECTURE HIGHLIGHTS

- Designed and deployed Retrieval-Augmented Generation (RAG) pipelines combining vector search and LLM reasoning for enterprise document retrieval.
- Built multi-agent orchestration systems using LangChain and LangGraph enabling coordinated task execution across specialized AI agents.
- Developed LLM-powered workflow automation for document analysis, proposal generation, and knowledge retrieval.
- Engineered scalable document ingestion pipelines including PDF parsing, semantic chunking, embedding generation, and vector indexing.
- Implemented LLM application backends using FastAPI with asynchronous task processing via Redis and Celery.

SELECTED PROJECTS

LearnVault

June 2026

(Golang, LLM APIs, Docker, Redis, Asynq, PostgreSQL, Next.js, Livekit, Caddy 2)

- Built LearnVault, a multi-tenant SaaS platform using Go, Next.js, PostgreSQL, and Docker, serving education, corporate training, and coaching organizations through isolated subdomain tenants.
- Developed AI-powered learning workflows including speech-to-text transcription, speaker diarization, automated summaries, quizzes, revision notes, and a RAG-based tutor using pgvector semantic search.
- Designed a subscription and credit billing system with regional pricing (NGN/USD/GBP), Flutterwave integration, atomic credit accounting, webhook verification, and comprehensive audit logging.
- Engineered a document-aware AI tutor with persistent chat sessions, conversation memory, semantic retrieval, and support for PDF, DOCX, PPTX, TXT, and CSV knowledge sources.
- Delivered live virtual learning, meeting, and coaching experiences, including recording pipelines, assessments, notifications, role-based access control, and a multi-tenant administration platform supporting educational institutions, corporate teams, and coaching organizations.

MailCheckr

March 2026

(Python, FastAPI, IMAP, LLM APIs, Docker, Celery, Redis, PostgreSQL, Next.js)

- Centralized multiple IMAP accounts so the client no longer switches between inboxes; saving daily context-switch time.
- Real-time fetch status and notifications accelerated response time to important messages and reduced missed items.
- LLM-based formatting and translation turned messy or foreign emails into readable summaries, enabling faster understanding and decision-making.
- Built-in triage actions (snooze, archive, move, spam, trash) streamlined workflow and reduced manual inbox management overhead.
- Dockerized deployment and admin tools simplified onboarding and maintenance, letting the client focus on work instead of operational setup.

InfiniMail (Private client project; proprietary code)

October 2025

(Python, FastAPI, Next.js, PostgreSQL, Celery, Redis, Docker, Gmail API, REST APIs, JWT Authentication)

- Built a full-stack email automation platform (FastAPI + Next.js + Docker) with AI-ready ingestion (document & CSV uploads) and scalable async workers (Celery) for background email generation and notifications.

- Implemented secure authentication/session management and role-based admin controls, including referral tracking, subscription billing (monthly/yearly), crypto & bank payment flows, and withdrawal processing with balance ledger logic.
- Engineered robust email lifecycle: campaign workflow, scheduling, delivery status tracking, and Gmail API integration with structured notification agents for payments and withdrawals.
- Designed performant REST API layer (pagination, filtering, validation) and optimized database interactions for transactional actions (transactions, balances, withdrawals) with clear audit logging.
- Integrated real-time user support (Tawk), admin tooling, and analytics-friendly data models; standardized UX components (history tables, tabbed dashboards) for maintainable, production-grade UI.

EDUCATION AND CERTIFICATIONS

Google Prompting Essentials CERTIFICATE	September 2025
Python and Statistics for Financial Analysis CERTIFICATE	May 2024
IBM AI Engineering (IBM) CERTIFICATE	April 2024
Google Advanced Data Analytics (Google) CERTIFICATE	April 2024
Google Data Analytics (Google) CERTIFICATE	October 2023
Nnamdi Azikiwe University BSc Soil Science and Land Resources Management	Anambra, Nigeria August 2017 – April 2023
Languages • English – C1 (Advanced, strong written and spoken)	